Creating Sprites

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Sprites are objects, with different properties like height, width, color, etc., and methods like moving right, left, up and down, jump, etc. Here er are looking to create an object in which user can control that object and move it forward, backward, up and down using arrow keys.

```
import pygame
import random
# Global variables.
COLOR = (255, 100, 98)
SURFACE_COLOR = (167,255,100)
WIDTH = 500
HEIGHT = 500
# Object class.
class Sprite(pygame.sprite.Sprite):
    def __init__(self, color, height, width):
        super().__init__()

        self.image = pygame.Surface([width, height])
        self.image.fill(SURFACE_COLOR)
        self.image.set_colorkey(COLOR)

        pygame.draw.rect(self.image, color, pygame.Rect(0,0,width, height))

        self.rect = self.image.get_rect()

pygame.init()
RED=(255,0,0)
size=(WIDTH, HEIGHT)
screen = pygame.display.set_mode(size)
pygame.display.set_caption("Creating Sprite")
all_sprite_list = pygame.sprite.Group()
object_ = Sprite(RED, 20, 30)
object_.rect.x = 200
object_.rect.y = 300
all_sprite_list.add(object_)
clock = pygame.time.Clock()
while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            quit()

    all_sprite_list.update()
    screen.fill(SURFACE_COLOR)
    all_sprite_list.draw(screen)
    pygame.display.flip()
    clock.tick(60)
```

How to control the sprite, like moving forward, backward, slow or accelerate, and some of the properties sprite should have. We will be adding event handlers to

our program to respond to keystrokes events, when the player uses the arrow keys on the keyboard we will call our pure method to move the object on the screen.

```
import pygame
import random
# Global variables.
COLOR = (255, 100, 98)
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WIDTH = 500
HEIGHT = 500
# Object class.
class Sprite(pygame.sprite.Sprite):
    def __init__(self, color, height, width):
        super().__init__()

        self.image = pygame.Surface([width, height])
        self.image.fill(SURFACE_COLOR)
        self.image.set_colorkey(COLOR)

        pygame.draw.rect(self.image, color, pygame.Rect(0,0,width, height))

        self.rect = self.image.get_rect()

    def moveRight(self, pixels):
        self.rect.x += pixels

    def moveLeft(self, pixels):
        self.rect.x -= pixels

    def moveForward(self, speed):
        self.rect.y += speed * speed/10

    def moveBack(slef, speed):
        slef. rect.y -= speed * speed/10

pygame.init()
RED=(255,0,0)
size=(WIDTH, HEIGHT)
screen = pygame.display.set_mode(size)
pygame.display.set_caption("Creating Sprite")
all_sprite_list = pygame.sprite.Group()
playerCar = Sprite(RED, 20, 30)
playerCar.rect.x = 200
playerCar.rect.y = 300
all_sprite_list.add(playerCar)
clock = pygame.time.Clock()
while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            quit()
    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        playerCar.moveLeft(10)
```
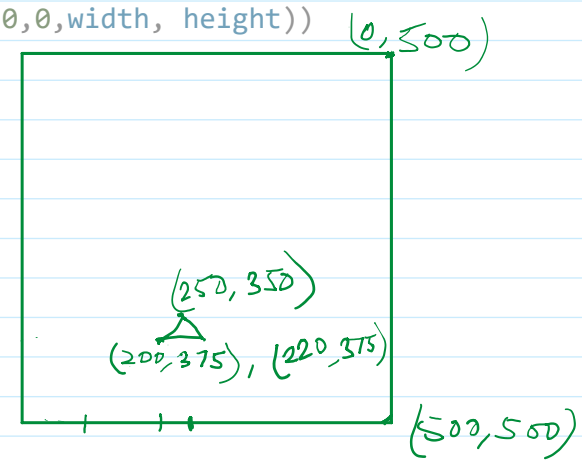
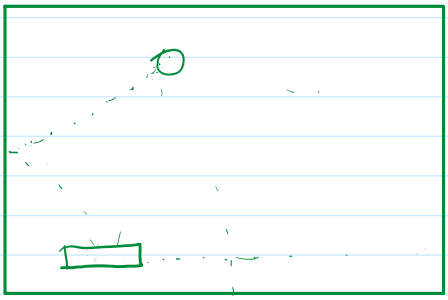



```
    if keys[pygame.K_RIGHT]:
        playerCar.moveRight(10)

    if keys[pygame.K_DOWN]:
        playerCar.moveForward(10)

    if keys[pygame.K_UP]:
        playerCar.moveBack(10)


    all_sprite_list.update()
    screen.fill(SURFACE_COLOR)
    all_sprite_list.draw(screen)
    pygame.display.flip()
    clock.tick(60)
```



Code for triangle sprite:

```
import pygame
import random
# Global variables.
COLOR = (255, 100, 98)
SURFACE_COLOR = (167,255,100)
WIDTH = 500
HEIGHT = 500
# Object class.
class Sprite(pygame.sprite.Sprite):
    def __init__(self, color, height, width):
        super().__init__()

        self.image = pygame.Surface([width, height])
        self.image.fill(SURFACE_COLOR)
        self.image.set_colorkey(COLOR)

        # pygame.draw.rect(self.image, color, pygame.Rect(0,0,width, height))
        # Define triangle points (pointing upwards)
        points = [
            (width // 2, 0),        # Top middle
            (0, height),            # Bottom left
            (width, height)         # Bottom right
        ]
        pygame.draw.polygon(self.image, color, points)

        self.rect = self.image.get_rect()

    def moveRight(self, pixels):
        self.rect.x += pixels

    def moveLeft(self, pixels):
        self.rect.x -= pixels

    def moveForward(self, speed):
        self.rect.y += speed * speed/10
```

```
    def moveBack(slef, speed):
        slef. rect.y -= speed * speed/10

pygame.init()
RED=(255,0,0)
size=(WIDTH, HEIGHT)
screen = pygame.display.set_mode(size)
pygame.display.set_caption("Creating Sprite")
all_sprite_list = pygame.sprite.Group()
playerCar = Sprite(RED, 20, 30)
playerCar.rect.x = 200
playerCar.rect.y = 300
all_sprite_list.add(playerCar)
clock = pygame.time.Clock()
while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            quit()
    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        playerCar.moveLeft(10)

    if keys[pygame.K_RIGHT]:
        playerCar.moveRight(10)

    if keys[pygame.K_DOWN]:
        playerCar.moveForward(10)

    if keys[pygame.K_UP]:
        playerCar.moveBack(10)


    all_sprite_list.update()
    screen.fill(SURFACE_COLOR)
    all_sprite_list.draw(screen)
    pygame.display.flip()
    clock.tick(60)
```

Home work

Practice Question 1: Basic Sprite Movement
Write a Pygame program to create a **rectangular sprite** of size 40 × 20 and color **blue.**
The sprite should:

- Move **left** when the LEFT arrow key is pressed
- Move **right** when the RIGHT arrow key is pressed
- Move **up** when the UP arrow key is pressed
- Move **down** when the DOWN arrow key is pressed

Ensure the sprite stays **within the screen boundaries (500 × 500 window).**

Practice Question 2: Speed Control (Acceleration & Deceleration)

Modify a sprite class so that:

- Pressing **UP arrow** increases the sprite's speed gradually (acceleration)
- Pressing **DOWN arrow** decreases the sprite's speed gradually (deceleration)
- LEFT and RIGHT arrows move the sprite horizontally using the current speed

Display the **current speed value** on the screen using pygame.font.

Practice Question 3: Triangle Sprite Rotation

Create a **triangle-shaped sprite** using pygame.draw.polygon().

Requirements:

- LEFT arrow → rotate the triangle **anticlockwise**
- RIGHT arrow → rotate the triangle **clockwise**
- UP arrow → move the triangle forward in the direction it is facing

(Hint: Use angles and pygame.transform.rotate().)

Practice Question 4: Multiple Sprites Control

Write a program that creates **two sprites**:

- Player 1 controlled using **arrow keys**
- Player 2 controlled using **W, A, S, D keys**

Each sprite should:

- Have a different color and shape
- Move independently
- Stop moving when reaching the window boundary

Practice Question 5: Collision Detection

Create a game where:

- One sprite is controlled using arrow keys
- One sprite moves automatically from top to bottom

If both sprites **collide**:

- Change the background color to **red**
- Display the message **"Collision Detected!"** on the screen

(Hint: Use pygame.sprite.collide_rect().)